Practical No.4

Database and Networking

1. Develop an Android application implementing SQLite database CRUD operations.
2. Develop an Android application to fetch and parse JSON data from a web service and display it in UI components.

1.Database Programming with SQLite in java language

AIM: To study and implement Database Programming with SQLite in java language

- Create SQLite Database
- Create Table
- Insert Data
- View Data
- Update Data
- Delete Data
- Display Output in TextView

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
 <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="20dp">
<EditText
            android:id="@+id/etName"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="Enter Name"/>
 <EditText
            android:id="@+id/etMarks"
            android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Enter Marks"
        android:inputType="number"/>
<Button
        android:id="@+id/btnInsert"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Insert"/>
 <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Data"/>
    <Button
        android:id="@+id/btnUpdate"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Update"/>
    <Button
        android:id="@+id/btnDelete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Delete"/>
    <TextView
        android:id="@+id/tvResult"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="16sp"
        android:paddingTop="20dp"/>
```

```
    </LinearLayout>
</ScrollView>
```

DatabaseHelper.java

```
package com.example.sqliteexample;
import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
public class DatabaseHelper extends SQLiteOpenHelper {
    private static final String DATABASE_NAME = "Student.db";
    private static final String TABLE_NAME = "student_table";
    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }
    @Override
    public void onCreate(SQLiteDatabase db) {
        db.execSQL("CREATE TABLE " + TABLE_NAME +
                " (ID INTEGER PRIMARY KEY AUTOINCREMENT, NAME TEXT, MARKS
INTEGER)");
    }
    @Override
    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
        db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
        onCreate(db);
    }
    public boolean insertData(String name, String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put("NAME", name);
        contentValues.put("MARKS", marks);
```

```
        long result = db.insert(TABLE_NAME, null, contentValues);
        return result != -1;
    }
    public Cursor getAllData() {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
    }
    public boolean updateData(String id, String name, String marks) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues contentValues = new ContentValues();
        contentValues.put("ID", id);
        contentValues.put("NAME", name);
        contentValues.put("MARKS", marks);
        db.update(TABLE_NAME, contentValues, "ID = ?", new String[]{id});
        return true;
    }
    public Integer deleteData(String id) {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.delete(TABLE_NAME, "ID = ?", new String[]{id});
    }
}
```

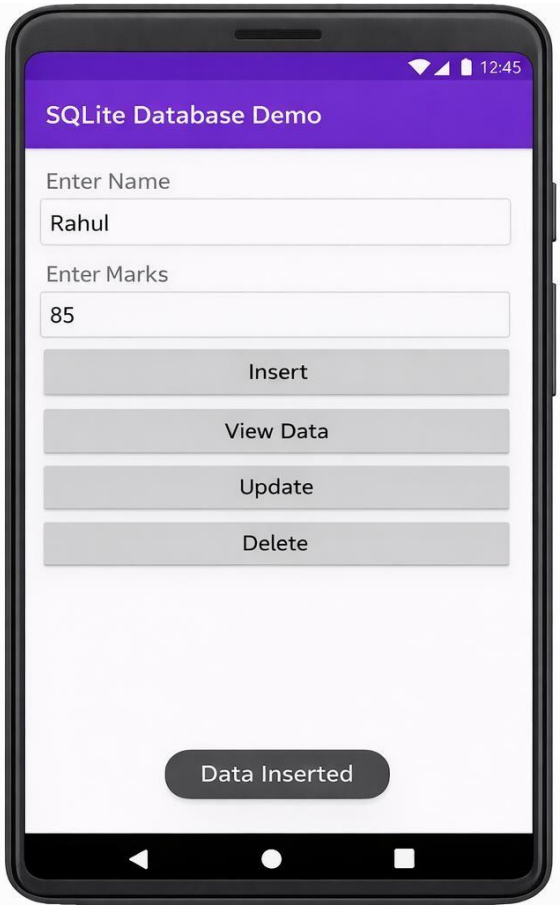
MainActivity.java

```
package com.example.sqliteexample;
import androidx.appcompat.app.AppCompatActivity;
import android.database.Cursor;
import  android.os.Bundle;
import android.view.View;
import android.widget.*;
```

```
public class MainActivity extends AppCompatActivity {
    EditText etName, etMarks;
    Button btnInsert, btnView, btnUpdate, btnDelete;
    TextView tvResult;
    DatabaseHelper myDb;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        myDb = new DatabaseHelper(this);
        etName = findViewById(R.id.etName);
        etMarks = findViewById(R.id.etMarks);
        btnInsert = findViewById(R.id.btnInsert);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);
        tvResult = findViewById(R.id.tvResult);
        btnInsert.setOnClickListener(view -> {
            boolean isInserted = myDb.insertData(
                    etName.getText().toString(),
                    etMarks.getText().toString());
            if (isInserted)
                Toast.makeText(this, "Data Inserted", Toast.LENGTH_SHORT).show();
            else
                Toast.makeText(this, "Data Not Inserted", Toast.LENGTH_SHORT).show();
        });
        btnView.setOnClickListener(view -> {
            Cursor res = myDb.getAllData();
            if (res.getCount() == 0) {
```

```
            tvResult.setText("No Data Found");
            return;
        }
        StringBuffer buffer = new StringBuffer();
        while (res.moveToNext()) {
            buffer.append("ID : " + res.getString(0) + "\n");
            buffer.append("Name : " + res.getString(1) + "\n");
            buffer.append("Marks : " + res.getString(2) + "\n\n");
        }
        tvResult.setText(buffer.toString());
    });
    btnUpdate.setOnClickListener(view -> {
        String id = "1"; // Example: updating ID 1
        boolean isUpdate = myDb.updateData(
                id,
                etName.getText().toString(),
                etMarks.getText().toString());
        if (isUpdate)
            Toast.makeText(this, "Data Updated", Toast.LENGTH_SHORT).show();
        else
            Toast.makeText(this, "Data Not Updated", Toast.LENGTH_SHORT).show();
    });
    btnDelete.setOnClickListener(view -> {
        String id = "1"; // Example: deleting ID 1
        Integer deletedRows = myDb.deleteData(id);
        if (deletedRows > 0)
            Toast.makeText(this, "Data Deleted", Toast.LENGTH_SHORT).show();
        else
            Toast.makeText(this, "Data Not Deleted", Toast.LENGTH_SHORT).show();
```

```
        });
    }
}
```


12:45
SQLite Database Demo
Enter Name
Rahul
Enter Marks
85
Insert
View Data
Update
Delete
Data Inserted


12:45
SQLite Database Demo
Insert
View Data
Update
Delete
ID : 1
Name : Rahul
Marks : 85